NORMAL SKIN DERMATOLOGY

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2025 Edition

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Normal Skin Dermatology

The Skin as a Window to Health

The skin is the largest organ of the human body, accounting for approximately 15% of total body weight and covering a surface area of roughly 1.5–2 m^2 in adults. It serves as a dynamic interface between the internal and external environments. Understanding its structure, the language of its lesions, and the systematic approach to dermatologic diagnosis forms the bedrock of clinical dermatology. This comprehensive guide provides a thorough exploration of skin anatomy, the classification of skin lesions, and the diagnostic framework used by dermatologists in everyday practice.

The skin is far more than a passive covering — it provides a living, self-renewing barrier against mechanical injury, ultraviolet radiation, microbial invasion, and chemical penetration. Through vasodilation, vasoconstriction, and eccrine sweat gland activity, it plays a central role in maintaining core body temperature. The stratum corneum and intercellular lipid matrix prevent excessive transepidermal water loss, maintaining hydration homeostasis. Critically, underlying systemic conditions — including autoimmune disorders, malignancies, endocrine diseases, and infections — can manifest with cutaneous signs before other organ involvement becomes apparent. Close inspection of the entire skin surface, nails, and mucous membranes should therefore be an integral component of every routine physical examination.

Skin diseases carry a significant psychosocial burden. Conditions such as psoriasis, acne, vitiligo, and eczema can profoundly affect self-esteem, social functioning, and mental health. The Dermatology Life Quality Index (DLQI) is a validated, 10-item questionnaire used to quantify the impact of skin disease on a patient's quality of life, with scores ranging from 0 (no impact) to 30 (extreme impact).

KEY TEACHING POINTS

The skin is the body's largest organ, accounting for ~15% of body weight and covering 1.5–2 m^2 in adults.

Primary functions: biological barrier, thermoregulation, prevention of water loss, immune surveillance, and sensory perception.

Always examine the entire skin, nails, and mucous membranes — cutaneous findings may be the first clue to an underlying internal disorder.

The DLQI is the standard validated tool for measuring quality-of-life impact of dermatologic conditions in clinical practice and research.

Skin Anatomy

Layers of the Skin

The skin is composed of three principal layers, each with distinct structural components and functions. A thorough understanding of skin anatomy is essential for interpreting the morphology and behaviour of skin lesions, planning surgical procedures, and understanding the pathophysiology of dermatologic disease.

1. Epidermis

The epidermis is the outermost layer of the skin and is composed predominantly of keratinocytes (squamous epithelial cells). It is avascular, receiving its nutrients by diffusion from the underlying dermal capillaries. The epidermis is rich in keratin, a structural protein that provides mechanical strength and waterproofing.

The epidermis is further subdivided into five strata from deepest to most superficial. The stratum basale (basal layer) consists of a single layer of columnar cells resting on the basement membrane; this is the germinative layer where keratinocyte proliferation occurs. Melanocytes, which produce melanin, are also located here. The stratum spinosum (prickle cell layer) contains several layers of polyhedral keratinocytes connected by desmosomes, giving a "spiny" appearance on histology. The stratum granulosum (granular layer) contains cells with keratohyalin granules that begin terminal differentiation. The stratum lucidum is a thin, translucent layer found only in thick skin (palms and soles). The stratum corneum (horny layer) is the outermost layer composed of anucleate, flattened corneocytes embedded in a lipid matrix; this is the primary barrier to water loss and environmental insult.

Other important epidermal cell types include melanocytes, which are dendritic cells located in the stratum basale that synthesize melanin pigment and transfer it to surrounding keratinocytes via melanosomes. Melanin determines skin colour and provides photoprotection. Langerhans cells are antigen-presenting dendritic cells distributed throughout the epidermis that form part of the cutaneous immune surveillance system. They capture antigens, migrate to regional lymph nodes, and present them to T lymphocytes. Merkel cells are mechanoreceptors found in the stratum basale, particularly abundant in fingertip skin, lips, and hair follicles; they mediate fine touch sensation.

The entire epidermis is approximately 0.1 mm thick on most body surfaces, though it ranges from 0.05 mm on the eyelids to over 1 mm on the palms and soles. The transit time for a keratinocyte to migrate from the basal layer to the surface (epidermal turnover time) is approximately 28–30 days in normal skin, but this is markedly accelerated in conditions such as psoriasis (3–5 days).

2. Dermis

The dermis lies beneath the epidermis, separated by the dermal-epidermal junction (basement membrane zone). It is a connective tissue layer that provides structural support, nourishment, and elasticity to the skin. The dermis is typically 1–4 mm thick, varying by body region (thickest on the back, thinnest on the eyelids).

The dermis contains connective tissue composed of collagen fibres (types I and III) that provide tensile strength, and elastin fibres that provide elasticity and resilience. The ground substance consists of glycosaminoglycans (including hyaluronic acid), proteoglycans, and glycoproteins. A superficial plexus (subpapillary plexus) and a deep plexus (reticular dermal plexus) connected by ascending arterioles and descending venules form the dermal vasculature. These blood vessels nourish the avascular epidermis via diffusion and play a critical role in thermoregulation and wound healing. A network of blind-ended lymphatic capillaries drains interstitial fluid and transports antigen-presenting cells to regional lymph nodes. Sensory nerve fibres (both myelinated Aδ fibres and unmyelinated C fibres) mediate touch, pressure, temperature, pain, and itch. Autonomic nerve fibres regulate eccrine sweat glands and vascular tone.

The dermis is subdivided into the papillary dermis (the superficial, thinner layer with loose connective tissue and dermal papillae that interdigitate with epidermal rete ridges) and the reticular dermis (the deeper, thicker layer with dense irregular connective tissue and thick collagen bundles).

3. Subcutis (Hypodermis)

The subcutis, or hypodermis, is the deepest layer of the skin, lying beneath the dermis. It is composed primarily of adipose tissue (fat lobules) separated by fibrous septa containing blood vessels, lymphatics, and nerves. The subcutis provides insulation, energy storage, mechanical cushioning, and anchorage of the skin to underlying structures such as fascia and muscle. The thickness of the subcutis varies considerably depending on body region, age, sex, and nutritional status.

KEY TEACHING POINTS

The epidermis is avascular and depends on dermal capillaries for nutrition; this has implications for wound healing and graft take.

Epidermal turnover time is ~28–30 days normally but accelerates to 3–5 days in psoriasis.

The dermal-epidermal junction (basement membrane zone) is the site of blister formation in diseases such as bullous pemphigoid and epidermolysis bullosa.

The subcutis provides insulation, cushioning, and energy storage; its thickness varies by body region and nutritional status.

Skin Appendages

The skin appendages are specialised structures derived from the epidermis that extend into the dermis or subcutis. They include hair follicles, sebaceous glands, eccrine sweat glands, apocrine sweat glands, and nails. Each plays a distinct role in skin function and homeostasis, and each can be the site of specific dermatologic pathology.

Hair Follicles

Hair follicles are distributed over the entire skin surface except for the palms, soles, lips, and glans penis. Each hair follicle consists of a hair shaft (composed of keratinised cells), an inner root sheath, an outer root sheath, and a hair bulb containing the dermal papilla (a vascularised connective tissue structure that nourishes the growing hair). The arrector pili muscle attaches to the follicle and causes hair erection ("goosebumps") in response to cold or sympathetic stimulation.

Hair growth occurs in a cyclical pattern with three phases. The anagen phase is the active growth phase lasting 2–6 years for scalp hair; the duration of anagen determines the maximum length of the hair. The catagen phase is a brief transitional phase (2–3 weeks) during which the lower two-thirds of the follicle undergoes apoptosis-driven involution. The telogen phase is the resting phase lasting approximately 3 months; during this phase, the hair is retained in the follicle as a "club hair" and is eventually shed. At any given time, approximately 85–90% of scalp hairs are in anagen, 1–2% in catagen, and 10–15% in telogen. Normal daily hair shedding is 50–100 hairs.

Sebaceous Glands

Sebaceous glands are multilobulated holocrine glands associated with hair follicles, forming the pilosebaceous unit. They are found in highest density on the face, scalp, and upper trunk. Sebaceous glands produce sebum, a lipid-rich mixture of triglycerides, wax esters, squalene, free fatty acids, and cholesterol. Sebum serves to lubricate the skin and hair, contribute to the acid mantle (skin surface pH ~4.5–5.5), and provide antimicrobial activity. Sebaceous gland activity is regulated primarily by androgens (testosterone and dihydrotestosterone), which is why sebaceous gland activity increases at puberty and why anti-androgen therapies can reduce sebum production.

Eccrine Sweat Glands

Eccrine sweat glands are simple, coiled tubular glands distributed over nearly the entire body surface, with the highest density on the palms, soles, and forehead. They produce a dilute, hypotonic secretion (sweat) composed primarily of water, sodium chloride, urea, ammonia, and lactate. Eccrine glands are innervated by sympathetic cholinergic nerve fibres and are the primary effectors of thermoregulatory sweating. Emotional stress can also stimulate eccrine secretion, particularly on the palms, soles, and axillae.

Apocrine Sweat Glands

Apocrine sweat glands are large, coiled glands found predominantly in the axillae, groin, areolae, perianal region, and external ear canal (ceruminous glands). They produce a viscous, lipid-rich, protein-rich secretion that is initially odourless but is metabolised by skin bacteria to produce body odour. Apocrine glands become active at puberty and are regulated by sympathetic adrenergic nerves. Apocrine glands do not play a significant role in thermoregulation.

Nails

The nail unit consists of the nail matrix (the germinative epithelium proximal to the lunula that produces the nail plate), the nail bed (the epithelium beneath the nail plate that contributes to nail adhesion and growth), the nail plate (the keratinised structure visible as the nail), the cuticle (eponychium), the hyponychium (the skin beneath the free edge of the nail), and the lateral nail folds. Fingernails grow at approximately 3 mm per month, and toenails at approximately 1 mm per month. Complete replacement of a fingernail takes 4–6 months; a toenail takes 12–18 months. Nail growth is influenced by age, season (faster in summer), pregnancy, and systemic disease.

KEY TEACHING POINTS

The pilosebaceous unit (hair follicle + sebaceous gland) is the key functional unit in acne pathogenesis.

Hair growth is cyclical: anagen (growth, 2–6 years), catogen (transition, 2–3 weeks), telogen (rest, ~3 months).

Normal daily hair shedding is 50–100 hairs; exceeding this suggests telogen effluvium or anagen effluvium.

Eccrine glands are the primary thermoregulatory glands; apocrine glands produce body odour and are hormonally regulated.

Fingernails grow ~3 mm/month; toenails ~1 mm/month — this is important when estimating the timing of systemic insults from nail changes.

Skin Blood Supply and Lymphatic Drainage

Vascular Anatomy

The skin has a rich and organised vascular supply that is essential for nutrition, thermoregulation, immune cell trafficking, and wound healing. Two horizontal plexuses are identified: the superficial (subpapillary) plexus located at the junction of the papillary and reticular dermis, and the deep (reticular) plexus located at the dermal-subcutaneous junction. These are connected by ascending arterioles that supply the dermal papillae, each of which contains a capillary loop that nourishes the overlying epidermis.

The subpapillary plexus gives rise to capillary loops within each dermal papilla. These loops have a descending arterial limb and an ascending venous limb and are the sites of nutrient exchange with the avascular epidermis. The rich vascular network of the skin allows it to participate in thermoregulation through vasodilation (heat dissipation) and vasoconstriction (heat conservation).

Lymphatic Drainage

Lymphatic capillaries in the dermis form an extensive network that drains interstitial fluid, transports immune cells (particularly Langerhans cells and dermal dendritic cells), and carries antigens to regional lymph nodes. The pattern of lymphatic drainage is clinically important in the staging and management of skin cancers, particularly melanoma and Merkel cell carcinoma. Sentinel lymph node biopsy relies on mapping the lymphatic drainage from a primary cutaneous tumour to the first draining (sentinel) lymph node.

KEY TEACHING POINTS

The two horizontal vascular plexuses (superficial and deep) are connected by ascending arterioles supplying dermal papillae.

Dermal capillary loops are the site of nutrient exchange with the avascular epidermis.

Lymphatic drainage patterns are critical for sentinel lymph node biopsy in melanoma staging.

Skin Innervation

The skin is richly innervated with both sensory and autonomic nerve fibres. Sensory innervation includes free nerve endings (mediating pain, temperature, and itch) and specialised nerve endings such as Meissner's corpuscles (light touch in dermal papillae), Pacinian corpuscles (deep pressure and vibration in the deep dermis and subcutis), and Merkel cell-neurite complexes (fine touch and texture discrimination).

Itch (pruritus) is mediated by a specialised subset of unmyelinated C-fibres that are distinct from pain fibres. These pruritoceptors respond to histamine, proteases, neuropeptides, and other pruritogens. The sensation of itch is conducted to the spinal cord via the spinothalamic tract and is subject to central modulation. Understanding the neurobiology of itch is essential for managing chronic pruritic conditions such as atopic dermatitis, urticaria, and prurigo nodularis.

Autonomic nerve fibres in the skin regulate eccrine sweat gland secretion (sympathetic cholinergic), apocrine gland secretion (sympathetic adrenergic), and vascular tone (sympathetic adrenergic). Dysregulation of cutaneous autonomic function contributes to conditions such as hyperhidrosis, rosacea, and Raynaud's phenomenon.

KEY TEACHING POINTS

Itch is mediated by specialised C-fibre pruritoceptors distinct from pain fibres — this explains why antihistamines help in some pruritic conditions but not all.

Meissner's corpuscles (dermal papillae) mediate light touch; Pacinian corpuscles (deep dermis/subcutis) mediate deep pressure and vibration.

Autonomic dysfunction in the skin contributes to hyperhidrosis, rosacea, and Raynaud's phenomenon.

Functions of the Skin

Barrier Function

The skin's barrier function is primarily mediated by the stratum corneum, which consists of corneocytes (anucleate, flattened keratinocytes) embedded in a lipid matrix composed of ceramides, cholesterol, and free fatty acids. This "bricks and mortar" model provides an effective barrier against transepidermal water loss (TEWL), chemical penetration, and microbial invasion. The acid mantle (skin surface pH ~4.5–5.5), maintained by sebum, sweat, and bacterial metabolism, further contributes to antimicrobial defence.

Disruption of the skin barrier — as occurs in atopic dermatitis, irritant contact dermatitis, and burns — leads to increased TEWL, dryness, and susceptibility to infection and sensitisation. Filaggrin gene mutations are a major genetic risk factor for atopic dermatitis and impair barrier function.

Thermoregulation

The skin is the body's primary thermoregulatory organ. Heat dissipation occurs through vasodilation of dermal blood vessels (increasing blood flow to the skin surface), eccrine sweat gland secretion (evaporative cooling), and radiation/convection. Heat conservation occurs through vasoconstriction, piloerection (minimal in humans), and behavioural responses. The cutaneous vascular bed can accommodate a wide range of blood flow — from as little as 1% of cardiac output in cold conditions to up to 60% during maximal vasodilation in heat stress.

Immune Function

The skin is an active immunological organ, often termed the skin-associated lymphoid tissue (SALT). Key immune cells in the skin include Langerhans cells (epidermal antigen-presenting cells), dermal dendritic cells, mast cells, T lymphocytes (both CD4+ and CD8+), and resident memory T cells. Keratinocytes themselves are not passive structural cells — they produce antimicrobial peptides (defensins, cathelicidins), cytokines (IL-1, IL-6, TNF-α, IL-8), and chemokines that actively participate in innate and adaptive immune responses.

Vitamin D Synthesis

The skin is the primary site of vitamin D3 (cholecalciferol) synthesis. Ultraviolet B (UVB) radiation (wavelength 290–315 nm) converts 7-dehydrocholesterol in the epidermis to pre-vitamin D3, which is then thermally isomerised to vitamin D3. Vitamin D3 is hydroxylated in the liver to 25-hydroxyvitamin D (calcidiol) and then in the kidney to the active form, 1,25-dihydroxyvitamin D (calcitriol). Adequate sun exposure (typically 5–15 minutes of midday sun on face and arms, 2–3 times per week) is sufficient for most individuals to maintain adequate vitamin D levels.

Sensory Function

The skin is the body's largest sensory organ, containing millions of nerve endings that detect touch, pressure, vibration, temperature, pain, and itch. This sensory input is essential for environmental awareness, protective reflexes, and social interaction (e.g., emotional touch).

KEY TEACHING POINTS

The "bricks and mortar" model of the stratum corneum: corneocytes (bricks) in a lipid matrix (mortar) of ceramides, cholesterol, and free fatty acids.

Filaggrin gene mutations are the strongest known genetic risk factor for atopic dermatitis.

The skin is the primary site of vitamin D3 synthesis via UVB conversion of 7-dehydrocholesterol.

Keratinocytes are active immune cells — they produce antimicrobial peptides, cytokines, and chemokines.

Cutaneous blood flow can range from 1% to 60% of cardiac output depending on thermoregulatory demands.

Skin Lesion Terminology

Primary Lesions

Accurate description of skin lesions using standardised terminology is the foundation of dermatologic diagnosis. Every skin lesion should be described using the agreed-upon lexicon of primary and secondary lesions. This common language allows clinicians to communicate precisely, formulate differential diagnoses, and recognise characteristic patterns of disease.

A macule is a flat, circumscribed area of colour change less than 1 cm in diameter. It is not palpable. Examples include freckles (ephelides), flat naevi, and the macular stage of a drug eruption. A patch is a flat, circumscribed area of colour change greater than 1 cm in diameter. Examples include vitiligo patches, port-wine stains, and café-au-lait macules (if >1 cm, they are sometimes called patches).

A papule is a small, solid, elevated lesion less than 1 cm in diameter. Examples include acne papules, lichen planus papules, and insect bite reactions. A plaque is a broad, flat-topped, elevated lesion greater than 1 cm in diameter, often formed by the confluence of papules. Examples include psoriasis plaques, lichen planus plaques, and mycosis fungoides plaques.

A nodule is a solid, palpable lesion greater than 1 cm in diameter, typically located in the dermis or subcutis. Examples include dermatofibromas, lipomas, and metastatic deposits. A tumour is a large, solid, palpable lesion, typically greater than 2 cm in diameter. The term does not imply malignancy — examples include dermatofibrosarcoma protuberans and large keloids.

A vesicle is a small, fluid-filled blister less than 1 cm in diameter. Examples include herpes simplex vesicles, dermatitis herpetiformis vesicles, and dyshidrotic eczema vesicles. A bulla is a large, fluid-filled blister greater than 1 cm in diameter. Examples include bullous pemphigoid bullae, burns blisters, and friction blisters.

A pustule is a small, circumscribed elevation of the skin containing purulent fluid (pus). Examples include acne pustules, folliculitis pustules, and impetigo pustules. A wheal is a transient, elevated, erythematous plaque caused by dermal oedema. Wheals are characteristically pruritic and blanch with pressure. Examples include urticarial wheals and dermatographism.

A cyst is a closed, sac-like structure filled with fluid, semi-solid, or solid material, lined by epithelium. Examples include epidermoid cysts (previously called sebaceous cysts), pilar cysts, and milia.

Secondary Lesions

Secondary lesions result from modification of primary lesions by scratching, infection, healing, or other external factors. A scale is a visible flake of dead epidermal cells (corneum) that accumulates on the skin surface. Examples include the silvery scales of psoriasis, the greasy scales of seborrheic dermatitis, and the fine pityriasis-like scales of pityriasis rosea.

A crust (scab) is a dried exudate of serum, blood, or pus on the skin surface. Examples include the honey-coloured crusts of impetigo and the haemorrhagic crusts of ruptured herpes simplex vesicles. An erosion is a superficial loss of epidermis that does not extend below the dermal-epidermal junction. Erosions heal without scarring. Examples include the erosions of pemphigus vulgaris and the superficial erosions left after rupture of vesicles.

An ulcer is a full-thickness loss of epidermis and dermis (and sometimes subcutis) that heals with scarring. Examples include venous ulcers, arterial ulcers, neuropathic ulcers (diabetic foot ulcers), and pyoderma gangrenosum. A fissure is a linear crack or split in the skin, often painful, that typically occurs in areas of thick skin or at skin folds. Examples include fissured hand eczema, angular cheilitis, and heel fissures.

Atrophy refers to thinning of the skin, which may involve the epidermis (appearing thin, wrinkled, and translucent, as in corticosteroid-induced atrophy), the dermis (appearing depressed, as in lichen sclerosus), or the subcutis (appearing as a depression, as in lipoatrophy). A scar is a permanent alteration of the skin resulting from damage to the dermis and subsequent fibrosis. Hypertrophic scars remain within the boundaries of the original wound; keloid scars extend beyond the original wound margins.

Lichenification is thickening of the skin with accentuation of normal skin markings, resulting from chronic rubbing or scratching. It is characteristic of chronic eczema and lichen simplex chronicus.

Special Lesions and Configurations

An excoriation is a linear erosion or ulcer caused by scratching. It is a hallmark of pruritic conditions such as atopic dermatitis, scabies, and dermatitis herpetiformis. A comedone is a dilated hair follicle plugging with keratin and sebum. Open comedones (blackheads) and closed comedones (whiteheads) are the hallmark primary lesions of acne vulgaris.

A burrow is a thin, linear, serpiginous tunnel in the epidermis created by the scabies mite (Sarcoptes scabiei). It is pathognomonic of scabies and is most commonly found in the web spaces of the fingers, wrists, and genitalia. A target lesion (iris lesion) is a lesion with three concentric zones of colour change — a central dark area, a pale oedematous ring, and an outer erythematous ring. It is the hallmark of erythema multiforme.

The configuration of skin lesions provides important diagnostic clues. Linear lesions suggest contact dermatitis (e.g., poison ivy), Koebner phenomenon (psoriasis, lichen planus), or external scratching. Annular (ring-shaped) lesions are seen in tinea corporis, granuloma annulare, erythema migrans (Lyme disease), and secondary syphilis. Reticulate (net-like) patterns are seen in livedo reticularis, erythema ab igne, and cutis marmorata. Herpetiform (grouped) vesicles are characteristic of herpes simplex and herpes zoster.

KEY TEACHING POINTS

Master primary lesion terminology — it is the language of dermatology and essential for accurate clinical description and communication.

Macule <1 cm (flat); patch >1 cm (flat); papule <1 cm (raised solid); plaque >1 cm (raised flat-topped).

Vesicle <1 cm (fluid-filled); bulla >1 cm (fluid-filled); pustule (pus-filled); wheal (transient, oedematous, pruritic).

Erosions heal without scarring (epidermal loss only); ulcers heal with scarring (dermal/subcutis loss).

Target lesions (3 concentric zones) = erythema multiforme; burrows = scabies; comedones = acne vulgaris.

Configuration matters: linear (contact/Koebner), annular (tinea/granuloma annulare), reticulate (livedo reticularis), herpetiform (herpes).

Approach to Dermatologic Diagnosis

History Taking

A systematic dermatologic history is essential for accurate diagnosis. The key components include the onset and duration of the lesion(s) — acute onset suggests urticaria, allergic contact dermatitis, or viral exanthem, while chronic lesions suggest psoriasis, eczema, or cutaneous T-cell lymphoma. The evolution of the lesion(s) should be documented — how the lesion started, how it has changed over time, and whether it has spread.

Symptoms are critical: pruritus is the most common symptom in dermatology and is prominent in atopic dermatitis, urticaria, scabies, and lichen planus. Pain suggests infection (cellulitis, abscess), herpes zoster, or vasculitis. Burning or stinging suggests contact dermatitis or neuropathic causes. The distribution of lesions provides diagnostic clues — sun-exposed areas suggest photosensitivity or porphyria; flexural areas suggest atopic dermatitis; extural areas suggest psoriasis.

A thorough drug history is essential, as drug eruptions are among the commonest causes of acute skin rashes. The temporal relationship between drug initiation and rash onset should be documented. A personal or family history of atopy (asthma, allergic rhinitis, atopic dermatitis) is relevant for atopic conditions. Occupational and hobby exposures are critical for contact dermatitis. A sexual history may be relevant for suspected sexually transmitted infections. Travel history is important for tropical infections and arthropod-borne diseases.

Physical Examination

Dermatologic examination should be performed in good lighting (natural daylight is ideal) with the entire skin surface exposed. The examination should include the skin, hair, nails, and mucous membranes. A dermatoscope (a handheld illuminated magnifying device with polarised or non-polarised light) is an essential tool for evaluating pigmented lesions and distinguishing benign from malignant lesions.

The systematic approach to describing skin lesions includes morphology (what type of primary and secondary lesion is present?), distribution (where on the body are the lesions located? symmetrical or asymmetrical? sun-exposed or covered?), configuration (are the lesions linear, annular, grouped, or reticulate?), colour (erythematous, violaceous, hyperpigmented, hypopigmented, skin-coloured?), border (well-demarcated or ill-defined?), and surface characteristics (smooth, scaly, crusted, verrucous?).

Diagnostic Tests

Skin biopsy is the most commonly performed diagnostic procedure in dermatology. A punch biopsy (typically 3–4 mm) provides a full-thickness sample of skin suitable for histopathological examination. Shave biopsies are used for superficial lesions. Excisional biopsies are preferred for suspected melanoma to allow assessment of Breslow thickness. Direct immunofluorescence (DIF) is essential for diagnosing autoimmune blistering diseases (pemphigus, pemphigoid, dermatitis herpetiformis).

Patch testing is the gold standard for diagnosing allergic contact dermatitis. Standard series of allergens are applied to the back under occlusive patches and read at 48 hours and 96–120 hours. A positive reaction (erythema, papules, vesicles) identifies the causative allergen. Potassium hydroxide (KOH) preparation is a rapid bedside test for fungal infections — skin scrapings are placed on a slide with 10–20% KOH and examined microscopically for hyphae or spores.

Tzanck smear involves scraping the base of a fresh vesicle and staining with Giemsa or Wright stain to look for multinucleated giant cells (herpes simplex or varicella-zoster virus) or acantholytic cells (pemphigus). Wood's lamp examination uses ultraviolet light (365 nm) to detect fluorescence patterns — coral-red fluorescence in erythrasma (Corynebacterium minutissimum), blue-green fluorescence in some dermatophyte infections, and white depigmentation in vitiligo.

Dermatoscopy (dermoscopy) uses a handheld device with magnification and polarised light to visualise subsurface structures not visible to the naked eye. It is essential for evaluating pigmented lesions and distinguishing benign naevi from melanoma. Key dermatoscopic features include pigment network, globules, streaks, blue-white structures, and atypical vascular patterns.

KEY TEACHING POINTS

Always take a thorough drug history — drug eruptions are among the commonest causes of acute skin rashes.

Examine the entire skin, nails, hair, and mucous membranes in good lighting (natural daylight preferred).

Dermatoscopy is essential for evaluating pigmented lesions — it significantly improves diagnostic accuracy for melanoma.

Punch biopsy is the workhorse of dermatopathology; direct immunofluorescence is required for autoimmune blistering diseases.

Patch testing is the gold standard for allergic contact dermatitis — read at 48h and 96–120h.

Skin Type and Photobiology

Fitzpatrick Skin Phototype Classification

The Fitzpatrick classification categorises skin into six phototypes based on the response to ultraviolet (UV) radiation, specifically the tendency to burn or tan. This classification is essential for assessing skin cancer risk, planning phototherapy, and predicting response to laser treatments.

Type I skin always burns and never tans — individuals typically have fair skin, blue or green eyes, and red or blonde hair. Type II skin usually burns and tans minimally — individuals typically have fair skin and blue eyes. Type III skin sometimes burns mildly and tans uniformly — individuals typically have darker white skin. Type IV skin rarely burns and tans easily — individuals typically have brown skin (Mediterranean, Asian, Hispanic). Type V skin very rarely burns and tans very easily — individuals typically have dark brown skin (Middle Eastern, South Asian, some African). Type VI skin never burns — individuals have deeply pigmented dark brown to black skin.

Ultraviolet Radiation and Skin Damage

Ultraviolet radiation is divided into UVA (320–400 nm), UVB (280–320 nm), and UVC (100–280 nm). UVC is absorbed by the ozone layer and does not reach the earth's surface. UVB is the primary cause of sunburn (erythema) and direct DNA damage through the formation of cyclobutane pyrimidine dimers (CPDs) and 6-4 photoproducts. UVB is also essential for vitamin D synthesis. UVA penetrates deeper into the dermis and causes indirect DNA damage through reactive oxygen species (ROS) generation. UVA is the primary driver of photoaging (dermal elastosis, wrinkling) and is responsible for many drug-induced photosensitivity reactions.

Cumulative UV exposure is the primary environmental risk factor for skin cancer. UVB-induced DNA damage, if not adequately repaired by nucleotide excision repair (NER) pathways, leads to characteristic C→T and CC→TT transition mutations at dipyrimidine sites. These mutations in tumour suppressor genes (particularly p53) and oncogenes drive the development of actinic keratoses, squamous cell carcinoma, and basal cell carcinoma. Melanoma is associated with intermittent intense UV exposure (sunburns), particularly in childhood.

KEY TEACHING POINTS

Fitzpatrick skin types range from I (always burns, never tans) to VI (never burns); this guides skin cancer risk assessment and phototherapy dosing.

UVB causes sunburn and direct DNA damage (cyclobutane pyrimidine dimers); UVA causes photoaging and indirect DNA damage via reactive oxygen species.

Cumulative UV exposure is the primary risk factor for non-melanoma skin cancer; intermittent intense exposure (sunburns) is the primary risk factor for melanoma.

Nucleotide excision repair (NER) deficiency (as in xeroderma pigmentosum) leads to extreme skin cancer susceptibility.

Wound Healing

Phases of Wound Healing

Wound healing is a complex, highly orchestrated process involving the coordinated interaction of platelets, inflammatory cells, fibroblasts, endothelial cells, keratinocytes, and extracellular matrix components. It proceeds through four overlapping phases.

Haemostasis occurs immediately after injury. Vasoconstriction reduces blood loss, and platelets adhere to exposed collagen, become activated, and form a platelet plug. The coagulation cascade generates fibrin, which cross-links to form a stable clot. The clot provides a provisional matrix for cell migration and serves as a reservoir of growth factors (PDGF, TGF-β, EGF).

The inflammatory phase begins within hours and lasts 2–5 days. Neutrophils are the first inflammatory cells to arrive, phagocytosing bacteria and debris. Monocytes migrate into the wound and differentiate into macrophages, which are the master regulators of wound healing. Macrophages phagocytose debris, produce growth factors (PDGF, TGF-β, VEGF, FGF), and orchestrate the transition from inflammation to repair. M1 macrophages (pro-inflammatory) predominate early; M2 macrophages (pro-reparative) predominate later.

The proliferative phase begins at approximately day 3–5 and lasts 2–3 weeks. Key events include angiogenesis (new blood vessel formation driven by VEGF and FGF), fibroplasia (fibroblast migration and proliferation, producing collagen type III and other extracellular matrix components), re-epithelialisation (keratinocyte migration and proliferation from wound edges and hair follicle stem cells), and granulation tissue formation (a vascularised, pink, moist tissue composed of new capillaries, fibroblasts, and loose extracellular matrix).

The remodelling (maturation) phase begins at approximately 3 weeks and can continue for months to years. Collagen type III is gradually replaced by stronger collagen type I. Collagen fibres are cross-linked and reorganised along lines of mechanical stress. The wound gains tensile strength progressively — at 3 weeks, the wound has approximately 20% of its original tensile strength; at 3 months, approximately 80%. Scar tissue never fully regains the strength of uninjured skin.

Factors Affecting Wound Healing

Local factors that impair wound healing include infection (the most common cause of delayed healing), ischaemia/poor blood supply, foreign bodies, repeated trauma or mechanical stress, and radiation damage. Systemic factors include diabetes mellitus (impaired neutrophil function, microvascular disease, reduced collagen synthesis), malnutrition (protein deficiency, vitamin C deficiency, zinc deficiency), immunosuppression (corticosteroids, chemotherapy, HIV), advanced age, smoking (vasoconstriction, reduced tissue oxygenation), and anaemia.

KEY TEACHING POINTS

Wound healing proceeds through four overlapping phases: haemostasis, inflammation, proliferation, and remodelling.

Macrophages are the master regulators of wound healing — they orchestrate the transition from inflammation to repair.

Wound tensile strength at 3 weeks is only ~20% of original; at 3 months, ~80% — this guides suture removal timing and activity restrictions.

Infection is the most common cause of delayed wound healing; diabetes and malnutrition are the most common systemic causes.

Vitamin C deficiency (scurvy) impairs collagen synthesis — clinical features include perifollicular haemorrhages, corkscrew hairs, and poor wound healing.

Skin Microbiome

The skin harbours a complex and diverse community of microorganisms — bacteria, fungi, viruses, and mites — collectively termed the skin microbiome. The composition of the skin microbiome varies by body site, reflecting the microenvironmental conditions of different skin regions. Moist sites (axillae, groin, web spaces) are dominated by Staphylococcus and Corynebacterium species. Sebaceous sites (face, chest, back) are dominated by Cutibacterium acnes (formerly Propionibacterium acnes). Dry sites (forearms, legs) harbour the greatest diversity of bacteria, including Betaproteobacteria and Flavobacteriales.

The skin microbiome plays a crucial role in skin health. Commensal bacteria compete with pathogenic organisms for nutrients and adhesion sites (colonisation resistance), produce antimicrobial peptides that inhibit pathogen growth, and educate the cutaneous immune system to distinguish between harmless commensals and dangerous pathogens. Disruption of the skin microbiome (dysbiosis) — through antibiotics, harsh cleansers, or underlying skin disease — can predispose to infection and inflammatory skin conditions.

KEY TEACHING POINTS

The skin microbiome varies by body site: moist areas (Staphylococcus, Corynebacterium), sebaceous areas (Cutibacterium acnes), dry areas (greatest diversity).

Commensal bacteria provide colonisation resistance against pathogens and educate the cutaneous immune system.

Dysbiosis of the skin microbiome is implicated in atopic dermatitis (Staphylococcus aureus overgrowth), acne (C. acnes strain imbalance), and rosacea (Demodex mite overgrowth).

Age-Related Skin Changes

Neonatal and Infant Skin

Neonatal skin differs from adult skin in several important ways. The stratum corneum is thinner and less mature, resulting in increased transepidermal water loss and greater susceptibility to chemical absorption and irritation. Neonatal skin has a higher surface pH (closer to neutral) compared to the acidic pH of adult skin, which reduces antimicrobial defence. Vernix caseosa, a white, cheese-like substance covering the skin of newborns, provides hydration, antimicrobial protection, and facilitates the postnatal acidification of the skin surface.

Common neonatal skin conditions include erythema toxicum neonatorum (benign, self-limiting erythematous papules and pustules with eosinophils on histology), neonatal acne (milia and comedones due to maternal androgen stimulation), and seborrheic dermatitis (cradle cap), which typically presents after the first few weeks of life.

Ageing Skin

Intrinsic (chronological) ageing is characterised by progressive thinning of the epidermis and dermis, decreased collagen production (approximately 1% per year after age 30), reduced elastin fibre quality, decreased sebaceous gland activity (leading to dryness), reduced sweat gland function, decreased immune function (immunosenescence), and impaired wound healing. The dermal-epidermal junction flattens with age, reducing the surface area for nutrient exchange and making the skin more susceptible to blistering.

Extrinsic ageing is primarily driven by chronic UV exposure (photoaging) and is characterised by deep wrinkles, solar elastosis (yellowish, thickened skin with degraded elastin), irregular pigmentation (solar lentigines, mottled hyperpigmentation), telangiectasias, and increased risk of skin cancer. Photoaged skin shows histological features of solar elastosis (degraded, disorganised elastic tissue in the upper dermis), increased matrix metalloproteinase activity, and reduced collagen synthesis. Smoking is a major independent risk factor for extrinsic skin ageing.

KEY TEACHING POINTS

Neonatal skin has a thinner stratum corneum, higher pH, and greater susceptibility to chemical absorption — important for topical medication dosing and safety.

Collagen decreases by ~1% per year after age 30 — this drives age-related skin thinning and wrinkling.

Photoaging (extrinsic ageing) is driven by chronic UV exposure and is characterised by solar elastosis, deep wrinkles, and irregular pigmentation.

Smoking is a major independent risk factor for premature skin ageing — the "smoker's face" phenotype includes periorbital wrinkles and greyish skin tone.

Psychodermatology

The relationship between the skin and the mind is bidirectional and clinically significant. Psychodermatology encompasses three categories of conditions. Psychophysiologic disorders are genuine dermatologic conditions that are exacerbated or triggered by psychological stress — examples include atopic dermatitis, psoriasis, acne, and urticaria. Stress activates the hypothalamic-pituitary-adrenal (HPA) axis and the sympathetic-adrenal-medullary (SAM) axis, releasing cortisol, catecholamines, and neuropeptides that modulate immune function and can trigger or worsen inflammatory skin disease.

Primary psychiatric disorders are psychiatric conditions that manifest with skin findings — examples include delusions of parasitosis (a fixed, false belief of skin infestation), dermatitis artefacta (self-inflicted skin lesions), and trichotillomania (compulsive hair pulling). Secondary psychiatric disorders are psychiatric consequences of having a skin disease — examples include depression, anxiety, social phobia, and body dysmorphic disorder in patients with visible skin conditions such as psoriasis, acne, vitiligo, and alopecia areata.

The nocebo effect is relevant in dermatology — patients who expect side effects from topical medications may experience burning, stinging, or worsening of their condition due to psychological expectation rather than pharmacological effect. Conversely, the placebo effect can produce measurable improvement in inflammatory skin conditions.

KEY TEACHING POINTS

Psychodermatology encompasses psychophysiologic disorders (stress-triggered skin disease), primary psychiatric disorders (psychiatric conditions with skin manifestations), and secondary psychiatric disorders (psychiatric consequences of skin disease).

Stress activates the HPA axis and SAM axis, releasing mediators that can trigger or worsen inflammatory skin conditions.

Always screen for psychological distress in patients with chronic skin disease — the DLQI captures both physical and psychosocial impact.

Delusions of parasitosis, dermatitis artefacta, and trichotillomania are primary psychiatric disorders that present to dermatologists.

Dermatologic Emergencies

Several dermatologic conditions constitute medical emergencies requiring urgent recognition and management. These conditions are high-yield for examinations and clinical practice.

Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN) are severe, life-threatening mucocutaneous reactions most commonly triggered by drugs. SJS involves <10% body surface area (BSA) detachment, SJS-TEN overlap involves 10–30% BSA, and TEN involves >30% BSA. The hallmark is widespread epidermal necrosis with full-thickness epidermal detachment, presenting as painful erythema, blistering, and Nikolsky sign (lateral pressure on normal-appearing skin causes epidermal detachment). Mucosal involvement (oral, ocular, genital) is nearly universal. Mortality ranges from 10% in SJS to 30–50% in TEN. Common causative drugs include allopurinol, sulfonamide antibiotics, anticonvulsants (carbamazepine, phenytoin, lamotrigine), and nevirapine.

Necrotising fasciitis is a rapidly progressive, life-threatening soft tissue infection that spreads along fascial planes. It may be polymicrobial (Type I) or monomicrobial (Type II, most commonly Group A Streptococcus). Early recognition is critical — the skin may show erythema, oedema, and pain that is disproportionate to clinical findings. Later features include bullae, skin necrosis, crepitus (gas-producing organisms), and systemic sepsis. The LRINEC (Laboratory Risk Indicator for Necrotising Fasciitis) score can aid early diagnosis. Treatment requires urgent surgical debridement and broad-spectrum antibiotics.

Erythroderma (exfoliative dermatitis) is generalised erythema and scaling affecting >90% of the BSA. It may be caused by psoriasis, eczema, drugs, cutaneous T-cell lymphoma (mycosis fungoides/Sézary syndrome), or pityriasis rubra pilaris. Complications include hypothermia (loss of thermoregulation), high-output cardiac failure (increased cutaneous blood flow), hypoalbuminaemia (protein loss through scale), and secondary infection. Management requires careful fluid and electrolyte balance, temperature monitoring, and treatment of the underlying cause.

Staphylococcal scalded skin syndrome (SSSS) is caused by exfoliative toxins (ETA, ETB) produced by Staphylococcus aureus, which cleave desmoglein-1 in the superficial epidermis. It predominantly affects neonates and children under 5 years. Features include widespread tender erythema, superficial blistering, and Nikolsky sign, with sparing of mucous membranes (distinguishing it from SJS/TEN). Treatment is with anti-staphylococcal antibiotics (flucloxacillin).

KEY TEACHING POINTS

SJS/TEN: drug-induced, epidermal detachment (SJS <10%, TEN >30% BSA), Nikolsky sign positive, mucosal involvement universal — STOP the offending drug immediately.

Necrotising fasciitis: pain disproportionate to findings is the hallmark — urgent surgical debridement is life-saving.

Erythroderma: >90% BSA involvement — complications include hypothermia, high-output cardiac failure, and hypoalbuminaemia.

SSSS: exfoliative toxins cleave desmoglein-1, affects children <5 years, mucosal sparing (unlike SJS/TEN).

Common Benign Skin Lesions

Seborrheic Keratoses

Seborrheic keratoses (SKs) are the most common benign epithelial skin tumours. They present as well-demarcated, waxy, "stuck-on" appearing papules or plaques, often with a verrucous surface. Colour ranges from light tan to dark brown. They are extremely common in individuals over 40 years of age and increase in number with age. The "sign of Leser-Trélat" — the sudden eruption of multiple seborrheic keratoses – may indicate an underlying malignancy (most commonly gastric adenocarcinoma) and warrants investigation.

Dermatofibromas

Dermatofibromas are common benign fibrous nodules, typically presenting as firm, dome-shaped papules or nodules, most commonly on the legs. They are usually skin-coloured to brown and characteristically dimple on lateral compression (the "dimple sign" or "Fitzpatrick's sign"). They are benign and require no treatment unless symptomatic.

Lipomas

Lipomas are the most common benign soft tissue tumours, composed of mature adipose tissue. They present as soft, mobile, painless subcutaneous nodules, most commonly on the trunk, neck, and proximal extremities. They are typically slow-growing and benign. Multiple lipomas may be associated with familial multiple lipomatosis or Dercum's disease (adiposis dolorosa).

Epidermoid Cysts

Epidermoid cysts (previously incorrectly called sebaceous cysts) are common dermal or subcutaneous cysts lined by stratified squamous epithelium and filled with keratinous material. They present as firm, mobile, dome-shaped nodules, often with a visible central punctum. They may become inflamed, infected, or rupture, causing significant pain and inflammation. Treatment is surgical excision with intact removal of the cyst wall to prevent recurrence.

Skin Tags (Acrochordons)

Skin tags (acrochordons) are soft, flesh-coloured or hyperpigmented pedunculated papules that occur most commonly in skin folds — the neck, axillae, groin, and inframammary regions. They are associated with obesity, insulin resistance, and type 2 diabetes mellitus. They are benign and can be removed by snip excision, cryotherapy, or electrocautery if symptomatic or for cosmetic reasons.

Cherry Angiomas

Cherry angiomas (Campbell de Morgan spots) are common benign vascular proliferations that present as small (1–5 mm), bright red to violaceous papules, most commonly on the trunk. They increase in number with age and are benign. They may be confused with petechiae or early melanoma (amelanotic melanoma), but they are easily blanched with diascopy and have a characteristic dermatoscopic appearance.

KEY TEACHING POINTS

Seborrheic keratoses: "stuck-on" appearance, waxy, verrucous — the sign of Leser-Trélat (sudden eruption) may indicate internal malignancy.

Dermatofibromas: firm nodule on legs, positive "dimple sign" on lateral compression — benign, no treatment needed.

Epidermoid cysts: central punctum, filled with keratin — complete excision of cyst wall required to prevent recurrence.

Skin tags: associated with obesity, insulin resistance, and type 2 diabetes — consider screening for metabolic syndrome.

Cherry angiomas: benign vascular papules, increase with age — distinguish from petechiae and amelanotic melanoma.

Pigmented Lesions

Benign Naevi (Moles)

Melanocytic naevi (moles) are benign proliferations of melanocytes. They are classified by timing as congenital (present at birth or within the first year of life) or acquired (developing during childhood and adolescence). Congenital naevi are classified by size: small (<1.5 cm), medium (1.5–19.9 cm), and large/giant (≥20 cm). Large congenital naevi carry an increased risk of melanoma (estimated lifetime risk 5–10%).

Acquired melanocytic naevi evolve through three histological stages. Junctional naevi have melanocyte nests at the dermal-epidermal junction and present as flat or slightly raised, dark brown macules. Compound naevi have melanocyte nests in both the epidermis and dermis and present as dome-shaped, pigmented papules. Intradermal naevi have melanocyte nests confined to the dermis and present as flesh-coloured or lightly pigmented, dome-shaped papules. This evolution from junctional → compound → intradermal is a normal maturation process.

The ABCDE criteria are used to assess pigmented lesions for possible melanoma: Asymmetry (one half does not match the other), Border irregularity (ragged, notched, or blurred edges), Colour change or variegation (multiple shades of brown, black, red, white, or blue), Diameter greater than 6 mm (the size of a pencil eraser), and Evolving (any change in size, shape, colour, or symptoms such as itching or bleeding).

Dysplastic Naevi

Dysplastic naevi (atypical moles) are acquired melanocytic naevi with architectural disorder and cytological atypia. They are larger than common naevi (often >5 mm), have irregular borders, and show colour variegation. They may be sporadic or familial (dysplastic naevi syndrome/familial atypical multiple mole melanoma syndrome — FAMMM). Patients with dysplastic naevi syndrome have an increased risk of melanoma and require regular dermatologic surveillance with total body photography and sequential dermatoscopy.

Freckles (Ephelides)

Freckles (ephelides) are small (1–3 mm), light brown macules that appear on sun-exposed skin, particularly in fair-skinned individuals (Fitzpatrick types I–II). They darken with sun exposure and fade in winter. Histologically, they show increased melanin in basal keratinocytes with a normal number of melanocytes. They are benign and require no treatment.

Lentigines

Lentigines are small, well-demarcated brown macules caused by an increased number of melanocytes at the dermal-epidermal junction (in contrast to freckles, which have a normal number of melanocytes with increased melanin). Solar lentigines (age spots, liver spots) occur on sun-exposed skin and are a marker of cumulative UV exposure. They do not fade in winter. Lentigo maligna is a form of melanoma in situ that arises on chronically sun-damaged skin, typically on the face of elderly patients.

KEY TEACHING POINTS

The ABCDE criteria (Asymmetry, Border, Colour, Diameter, Evolving) are the standard screening tool for melanoma.

Melanocytic naevi mature from junctional → compound → intradermal; this is a normal process, not malignant transformation.

Large congenital naevi (≥20 cm) carry a 5–10% lifetime risk of melanoma.

Dysplastic naevi syndrome (FAMMM) requires regular surveillance — these patients have a markedly increased melanoma risk.

Solar lentigines = marker of cumulative UV damage; lentigo maligna = melanoma in situ on sun-damaged skin.

Skin Cancer: An Overview

Basal Cell Carcinoma

Basal cell carcinoma (BCC) is the most common cancer in humans, accounting for approximately 80% of all non-melanoma skin cancers. It arises from basal keratinocytes of the epidermis and is locally invasive but rarely metastasises (metastatic rate <0.1%). Risk factors include chronic UV exposure (particularly intermittent intense exposure), fair skin (Fitzpatrick types I–II), immunosuppression, and Gorlin syndrome (naevoid basal cell carcinoma syndrome, caused by PTCH1 gene mutations).

The nodular subtype is the most common, presenting as a pearly, translucent papule or nodule with telangiectasias on the surface, often with central ulceration (rodent ulcer). The superficial subtype presents as a scaly, erythematous plaque, most commonly on the trunk. The morphoeic (sclerosing) subtype presents as a scar-like plaque with ill-defined borders and has the highest recurrence rate. Treatment options include surgical excision, Mohs micrographic surgery (for high-risk sites or morphoeic BCC), curettage and cautery, cryotherapy, photodynamic therapy, and topical imiquimod or 5-fluorouracil.

Squamous Cell Carcinoma

Squamous cell carcinoma (SCC) is the second most common skin cancer, accounting for approximately 20% of non-melanoma skin cancers. It arises from keratinocytes of the epidermis and has a higher metastatic potential than BCC (metastatic rate 2–6%, higher in immunosuppressed patients and in SCCs arising in chronic wounds or scars — Marjolin's ulcer). Risk factors include cumulative UV exposure, immunosuppression, chronic wounds/scars, HPV infection, and actinic keratoses (premalignant precursors).

SCC typically presents as a firm, erythematous, scaly or crusted nodule or plaque, often arising on sun-exposed skin. Keratoacanthoma is a rapidly growing variant that may be a well-differentiated SCC or a separate entity — it presents as a dome-shaped nodule with a central keratin plug and may regress spontaneously but is usually treated surgically. Treatment is primarily surgical excision with adequate margins. High-risk SCCs may require sentinel lymph node biopsy and adjuvant radiotherapy.

Melanoma

Melanoma is the most dangerous skin cancer, accounting for the majority of skin cancer deaths despite representing only ~5% of all skin cancers. It arises from melanocytes and has a high propensity for lymphatic and haematogenous metastasis. Risk factors include intermittent intense UV exposure (sunburns), particularly in childhood, fair skin, multiple naevi (>50), dysplastic naevi, family history of melanoma, and CDKN2A gene mutations (familial melanoma).

The four major subtypes are superficial spreading melanoma (the most common subtype, ~70%, characterised by a prolonged radial growth phase), nodular melanoma (~15–30%, characterised by early vertical growth and worse prognosis), lentigo maligna melanoma (arises from lentigo maligna on chronically sun-damaged skin, typically on the face of elderly patients), and acral lentiginous melanoma (the most common subtype in dark-skinned individuals, occurring on palms, soles, and nail beds — subungual melanoma presents as longitudinal melanonychia with Hutchinson's sign — pigmentation of the proximal nail fold).

Breslow thickness (measured in mm from the granular layer to the deepest tumour cell) is the single most important prognostic factor. The TNM staging system incorporates Breslow thickness, ulceration, mitotic rate, lymph node involvement, and distant metastasis. Sentinel lymph node biopsy is recommended for melanomas ≥1.0 mm Breslow thickness or ≥0.8 mm with ulceration. Adjuvant therapy with immune checkpoint inhibitors (anti-PD-1: nivolumab, pembrolizumab) or targeted therapy (BRAF/MEK inhibitors for BRAF-mutant melanoma) has significantly improved outcomes in high-risk and metastatic disease.

KEY TEACHING POINTS

BCC: most common skin cancer, locally invasive, rarely metastasises — nodular subtype is most common (pearly, telangiectatic, ulcerated).

SCC: second most common, higher metastatic potential than BCC — actinic keratoses are premalignant precursors; Marjolin's ulcer = SCC in chronic wound/scar.

Melanoma: most dangerous skin cancer — Breslow thickness is the most important prognostic factor.

Acral lentiginous melanoma is the most common subtype in dark-skinned individuals — check palms, soles, and nail beds.

Hutchinson's sign (pigmentation of the proximal nail fold) suggests subungual melanoma.

Sentinel lymph node biopsy is recommended for melanomas ≥1.0 mm Breslow thickness.

Dermatologic Pharmacology

Topical Corticosteroids

Topical corticosteroids are the most commonly prescribed medications in dermatology. They exert anti-inflammatory, immunosuppressive, antiproliferative, and vasoconstrictive effects. They are classified by potency into seven groups, from ultra-potent (Group I: clobetasol propionate 0.05%) to least potent (Group VII: hydrocortisone 1%).

Ultra-potent and potent steroids (Groups I–III) are used for short-term treatment of severe inflammatory dermatoses on thick skin (palms, soles, scalp). Medium-potency steroids (Groups IV–V) are used for moderate dermatoses on the trunk and limbs. Low-potency steroids (Groups VI–VII) are used for sensitive areas (face, groin, axillae, skin folds) and for prolonged use in children.

Side effects of topical corticosteroids include skin atrophy (thinning, striae, telangiectasias), perioral dermatitis, steroid-induced rosacea, rebound flare on withdrawal, tachyphylaxis (tolerance with prolonged use), and adrenal suppression with prolonged use of potent steroids over large body surface areas. The "finger-tip unit" (FTU) is a practical measure for prescribing topical medications — one FTU (approximately 0.5 g of cream ointment from a standard 5 mm nozzle tube) is sufficient to treat an area of skin equivalent to two adult handprints.

Other Topical Agents

Topical calcineurin inhibitors (tacrolimus 0.03%/0.1% ointment, pimecrolimus 1% cream) are steroid-sparing agents used for atopic dermatitis, particularly on the face and skin folds where steroid side effects are a concern. They do not cause skin atrophy. Topical retinoids (tretinoin, adapalene, tazarotene) are vitamin A derivatives used for acne, psoriasis, and photoaging. They work by normalising keratinocyte differentiation and reducing follicular plugging.

Topical antibiotics (clindamycin 1%, erythromycin 2%) are used for acne vulgaris, often in combination with benzoyl peroxide to reduce antibiotic resistance. Benzoyl peroxide is a topical antimicrobial and comedolytic agent used for acne — it is available over the counter in 2.5–10% concentrations. Topical antifungals (azole agents: ketoconazole, clotrimazole, miconazole; allylamines: terbinafine) are used for superficial fungal infections (dermatophytosis, candidiasis, pityriasis versicolor).

Systemic Agents

Systemic corticosteroids (prednisolone) are used for severe inflammatory and autoimmune dermatoses, including pemphigus vulgaris, bullous pemphigoid, severe drug reactions, and erythroderma. They carry significant side effects with prolonged use, including osteoporosis, diabetes, adrenal suppression, and increased infection risk.

Methotrexate is an antimetabolite (folate antagonist) used for psoriasis, atopic dermatitis, and autoimmune blistering diseases. It is given as a weekly dose (7.5–25 mg/week) with folic acid supplementation to reduce side effects. Monitoring includes full blood count, liver function tests, and renal function. Hepatotoxicity is a concern with cumulative doses — liver biopsy or FibroScan may be indicated.

Cyclosporine is a calcineurin inhibitor used for severe psoriasis and atopic dermatitis. It acts rapidly but is limited by nephrotoxicity and hypertension with prolonged use. Acitretin is an oral retinoid used for psoriasis, particularly pustular and erythrodermic forms. It is teratogenic (contraception required for 3 years after cessation in women of childbearing age) and can cause hyperlipidaemia and hepatotoxicity.

Biologic therapies have revolutionised the treatment of moderate-to-severe psoriasis and other inflammatory skin diseases. Anti-TNF-α agents (infliximab, adalimumab, etanercept), anti-IL-17 agents (secukinumab, ixekizumab), anti-IL-23 agents (guselkumab, risankizumab), and anti-IL-12/23 agent (ustekinumab) are all used in psoriasis. Dupilumab (anti-IL-4Rα) is used for atopic dermatitis. These agents are generally well-tolerated but require monitoring for infections (particularly tuberculosis reactivation with anti-TNF-α agents).

KEY TEACHING POINTS

Topical corticosteroids are classified into 7 potency groups — use the lowest effective potency for the shortest duration.

Finger-tip unit (FTU) is a practical prescribing measure: 1 FTU ≈ 0.5 g, treats 2 handprints of skin.

Topical calcineurin inhibitors (tacrolimus, pimecrolimus) are steroid-sparing — no skin atrophy, safe for face and skin folds.

Methotrexate: weekly dosing with folic acid supplementation; monitor FBC, LFTs, renal function.

Biologics have transformed psoriasis treatment — screen for latent TB before starting anti-TNF-α therapy.

Acitretin is teratogenic — contraception required for 3 years after stopping in women of childbearing age.

Practice Questions

Q: A 45-year-old man presents with a 6-month history of a slowly growing, painless, pearly nodule on his nose with visible telangiectasias on the surface. The lesion has a central ulceration. What is the most likely diagnosis?

A. Squamous cell carcinoma

B. Basal cell carcinoma

C. Melanoma

D. Seborrheic keratosis

E. Actinic keratosis

Answer: B

Explanation: The classic presentation of nodular basal cell carcinoma is a pearly, translucent papule or nodule with surface telangiectasias and central ulceration (rodent ulcer), typically on sun-exposed skin. BCC is the most common skin cancer and is locally invasive but rarely metastasises. SCC typically presents as a scaly or crusted nodule, melanoma would show pigmentation and ABCDE features, seborrheic keratosis has a "stuck-on" waxy appearance, and actinic keratosis is a rough, scaly macule or papule.

Q: A 32-year-old woman presents with a 2-day history of widespread, painful, erythematous skin with blistering and epidermal detachment involving 35% of her body surface area. She recently started allopurinol for gout. Nikolsky sign is positive. Oral and conjunctival mucosa are involved. What is the most likely diagnosis?

A. Staphylococcal scalded skin syndrome

B. Bullous pemphigoid

C. Toxic epidermal necrolysis

D. Erythema multiforme major

E. Pemphigus vulgaris

Answer: C

Explanation: Toxic epidermal necrolysis (TEN) is defined as epidermal detachment involving >30% BSA. It is most commonly drug-induced (allopurinol, sulfonamides, anticonvulsants). Nikolsky sign is positive, and mucosal involvement is nearly universal. SSSS spares mucous membranes and occurs predominantly in children. Bullous pemphigoid and pemphigus vulgaris are autoimmune blistering diseases with a more subacute course. Erythema multiforme major involves <10% BSA detachment with target lesions.

Q: A 60-year-old farmer presents with a 3-month history of a rapidly growing, firm, erythematous nodule on the left ear with a central keratin plug. The lesion appeared over 4 weeks and has been slowly enlarging. What is the most likely diagnosis?

A. Basal cell carcinoma

B. Actinic keratosis

C. Keratoacanthoma

D. Seborrheic keratosis

E. Dermatofibroma

Answer: C

Explanation: Keratoacanthoma presents as a rapidly growing, dome-shaped nodule with a central keratin plug, typically on sun-exposed skin. It may represent a well-differentiated SCC or a separate entity. It can regress spontaneously but is usually treated surgically. BCC has a pearly, telangiectatic appearance; actinic keratosis is a rough scaly macule; seborrheic keratosis has a "stuck-on" waxy appearance; dermatofibroma is a firm nodule with a positive dimple sign.

Q: A 28-year-old woman presents with multiple, small, firm, brown papules on her shins that dimple on lateral compression. The lesions are asymptomatic and have been present for several months. What is the most likely diagnosis?

A. Lipomas

B. Epidermoid cysts

C. Dermatofibromas

D. Cherry angiomas

E. Seborrheic keratoses

Answer: C

Explanation: Dermatofibromas are common benign fibrous nodules that characteristically occur on the legs and show a positive "dimple sign" (Fitzpatrick's sign) on lateral compression. They are firm, skin-coloured to brown, and benign. Lipomas are soft and mobile; epidermoid cysts have a central punctum; cherry angiomas are bright red vascular papules; seborrheic keratoses have a "stuck-on" waxy appearance.

Q: A 55-year-old man presents with a 4-month history of an irregularly shaped, darkly pigmented lesion on his back. The lesion is 8 mm in diameter, has irregular borders, and shows multiple shades of brown and black. It has been gradually enlarging. What is the most appropriate next step in management?

A. Reassurance and review in 3 months

B. Shave biopsy

C. Excisional biopsy with 2 mm margins

D. Topical imiquimod

E. Cryotherapy

Answer: C

Explanation: This lesion has multiple ABCDE features (Asymmetry, Border irregularity, Colour variegation, Diameter >6 mm, Evolving) suspicious for melanoma. The appropriate management is a full-thickness excisional biopsy with 2 mm margins to allow accurate assessment of Breslow thickness, which is the most important prognostic factor. Shave biopsy is inappropriate for suspected melanoma as it may not allow accurate depth measurement. Reassurance, topical therapy, and cryotherapy are not appropriate for a lesion suspicious for melanoma.

Q: A 70-year-old woman presents with a sudden eruption of multiple pruritic, erythematous papules and plaques over her trunk and limbs. On examination, the lesions are waxy and have a "stuck-on" appearance. She reports recent unintentional weight loss. What is the most important investigation?

A. Skin biopsy

B. CT scan of the abdomen

C. Patch testing

D. Serum IgE levels

E. KOH preparation

Answer: B

Explanation: The sudden eruption of multiple seborrheic keratoses is the sign of Leser-Trélat, which is a paraneoplastic phenomenon most commonly associated with gastric adenocarcinoma. The patient's unintentional weight loss further supports an underlying malignancy. CT scan of the abdomen (or upper GI endoscopy) is the most important investigation to identify the underlying malignancy. While skin biopsy would confirm the diagnosis of seborrheic keratoses, the priority is to investigate for internal malignancy.

Q: A 35-year-old man with a history of atopic dermatitis presents with a flare of severe eczema on his face and neck. He has been using a potent topical corticosteroid (betamethasone valerate 0.1%) for 6 weeks and now reports thinning of the skin and visible blood vessels. What is the most appropriate change in management?

A. Switch to an ultra-potent topical corticosteroid

B. Switch to a topical calcineurin inhibitor (tacrolimus ointment)

C. Start oral prednisolone

D. Start methotrexate

E. Continue the same treatment and add emollients

Answer: B

Explanation: The patient has developed skin atrophy and telangiectasias — side effects of prolonged potent topical corticosteroid use on the face. The most appropriate management is to switch to a topical calcineurin inhibitor (tacrolimus ointment), which is a steroid-sparing agent that does not cause skin atrophy and is safe for use on the face and skin folds. Ultra-potent steroids would worsen the atrophy. Systemic therapy is not indicated for a localised flare. Continuing the same treatment would perpetuate the side effects.

Q: A 50-year-old woman presents with a 1-week history of widespread, painful, erythematous skin with scaling affecting 95% of her body surface area. She has a long history of psoriasis. She is hypothermic and tachycardic. What is the most immediate concern?

A. Secondary bacterial infection

B. High-output cardiac failure

C. Hypoalbuminaemia

D. Thermoregulatory failure

E. All of the above

Answer: E

Explanation: This patient has erythroderma (exfoliative dermatitis) secondary to psoriasis, affecting >95% BSA. Erythroderma is a dermatologic emergency with multiple potential complications: thermoregulatory failure (hypothermia due to loss of vasomotor control), high-output cardiac failure (due to massively increased cutaneous blood flow), hypoalbuminaemia (due to protein loss through scale), and secondary infection (due to barrier disruption). All of these complications require immediate assessment and management, including careful fluid and electrolyte balance, temperature monitoring, and treatment of the underlying cause.